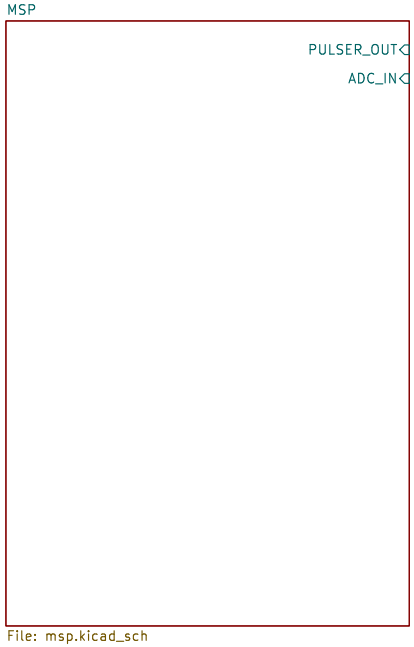
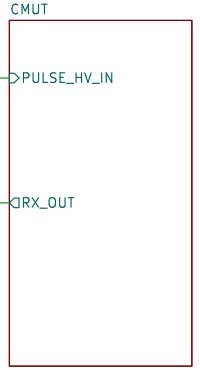














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Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.5		Id: 1/8

U1

MSP430FR5043IRGCT

AVCC1 1 — AVCC1
DVCC1 18 — DVCC1
DVCC3 49 — DVCC3
PVCC 56 — PVCC
PVCC 57 — PVCC
VGA_GAIN_RC_EN 3 — P1.0_UCA1CLK_TA1.0_A0_C0_VREF−_VeREF−
VGA_GAIN_CS 4 — P1.1_UCA1STE_TA4.0_A1_C1_VREF+_VeREF+
VGA_PWR_EN 19 — P1.2_UCA1SIMO_UCA1TXD_TA1.0_A8_C8
PREAMP_EN 20 — P1.3_UCA1SOMI_UCA1RXD_TA1.1_A9_C9
25 — P1.4_TB0.4_UCB0STE_A2_C2
PREAMP_PWR_EN 26 — P1.5_TB0.5_UCB0CLK_A3_C3
23 — P1.6_UCA3STE_UCB0SIMO_UCB0SDA
UUPS_TRIG_IN 24 — P1.7_USSTRG_UCA3CLK_UCB0SOMI_UCB0SCL
21 — P2.0_UCA1CLK_UCA3SIMO_UCA3TXD
22 — P2.1_UCA1STE_UCA3SOMI_UCA3RXD
VGA_GAIN_RC_SINK 2 — P2.2_COUT_UCA0CLK_A14_C14
LP_FILT_EN 27 — P3.1_TA1CLK_TB0.1_MTIF_OUT_IN
HV_POS_EN 50 — P3.3_MCLK_TB0.3_XPB1
HV_NEG_EN 51 — P3.4_SMCLK_DMAE0
HV_PWR_EN 52 — P3.5_ACLK_COUT
28 — P4.0_RTCCLK_TA4.1_MTIF_PIN_EN
LED_MSP430 29 — P4.1_UCA0CLK_TB0.4_UCA3SOMI_UCA3RXD
30 — P4.2_UCA0STE_TB0.5_UCA3SIMO_UCA3TXD
31 — P4.3_UCA0SIMO_UCA0TXD
32 — P4.4_UCA0SOMI_UCA0RXD
NRF_SPI.MOSI 34 — P5.0_TB0.0_UCA2SIMO_UCA2TXD
NRF_SPI.MISO 35 — P5.1_TB0.1_UCA2SOMI_UCA2RXD
NRF_SPI.CLK 36 — P5.2_TB0.2_UAC2CLK
NRF_SPI_CS 37 — P5.3_TB0.3_UCA2STE
VGA_SPI.CLK 38 — P5.4_TA0.0_UCB1CLK_TA4.0
VGA_SPI.MOSI 39 — P5.5_TA4.1_UCB1SIMO_UCB1SDA
VGA_SPI.MISO 40 — P5.6_TB0OUTH_UCB1SOMI_UCB1SCL
LINK_READY 41 — P5.7_TA0.2_UCB1STE
DATA_READY 42 — P6.0_TA0CLK_COUT
PULS_HIZ_IN 47 — P6.2_TB0CLK
SPARE_PIN 43 — P6.4_MCLK
UUPS_TRIG_IN 44 — P6.5_SMCLK
TRIG_OUT 45 — P6.6_ACLK
46 — P7.0_TA1.0_TA1.2_XPB0

PJ.0_UAC2CLK_SRSCG1_DMAE0_C10_TDO 13
PJ.1_UCA2STE_SRSCG0_TA4CLK_C11_TDI_TCLK 14
PJ.2_UCA2SIMO_UCA2TXD_SROSCOFF_TB0OUTH_C12_TMS 15
PJ.3_UCA2SOMI_UCA2RXD_SRCPUOFF_TB0.6_C13_TCK 16
PJ.4_LFXIN 6 — LFXIN
PJ.5_LFXOUT 7 — LFXOUT
PJ.6_HFXIN_USSXT_BOUT 9 — HFXIN
PJ.7_HFXOUT 10 — HFXOUT
CH0_IN 60 — CH0_IN
CH0_OUT 59 — CH0_OUT
CH1_IN 53
CH1_OUT 54
USSXTIN 62 — USSXTIN
TEST_SBWCLK 11 — JTAG_TEST
RST__NMI_SBWTDIO_N 12 — JTAG_RESET_N
USSXTOUT 63 — USSXTOUT
EP 65 — GND
AVSS1 64
AVSS2 5
AVSS3 8 — GND
AVSS4 61
DVSS1 17
DVSS2 33
DVSS3 48
PVSS 55 — GND
PVSS 58

2 VGA_GAIN_CS — VGA_GAIN_CS
2 VGA_GAIN_RC_EN — VGA_GAIN_RC_EN
2 VGA_PWR_EN — VGA_PWR_EN
2 PREAMP_EN — PREAMP_EN
2 PREAMP_PWR_EN — PREAMP_PWR_EN
2 VGA_GAIN_RC_SINK — VGA_GAIN_RC_SINK
2 LP_FILT_EN — LP_FILT_EN
2,4 HV_POS_EN — HV_POS_EN
2,4 HV_NEG_EN — HV_NEG_EN
2,4 HV_PWR_EN — HV_PWR_EN
2 NRF_SPI{SCLK, MOSI, MISO} — NRF_SPI.MISO, NRF_SPI.CLK, NRF_SPI.MOSI
2 NRF_MSP_CS — NRF_SPI_CS
2,3 VGA_SPI{CLK MOSI MISO} — VGA_SPI.MISO, VGA_SPI.CLK, VGA_SPI.MOSI
2 NRF_LINK_READY — LINK_READY
2 NRF_DATA_READY — DATA_READY
2 SPARE_PIN — SPARE_PIN
2 PULSER_HIZ_IN — PULS_HIZ_IN

C1 1n — ADC_IN
R2 1M — GND
R3 1M — GND
PULSER_OUT

TRIG_OUT — TP1 TestPoint

JTAG−SPY−wire

J1 Conn_01x04
4 — GND
3 — JTAG_TEST
2 — JTAG_RESET_N
1 — +VCC_MAP430
Stecker muss noch ausgewählt werden.
+VCC_MAP430
R1 47k
JTAG_RESET_N
C2 1n
GND

bypass caps and power supply filters

+VCC_MSP430
L1 600Ohm
C3 47u
C4 47u
C5 1n
C6 1n
GND
GND
PVCC

+VCC_MSP430
L2 600Ohm
C7 2.2u
C10 0.1u
AVCC1
GND

+VCC_MSP430
L3 600Ohm
C8 2.2u
C11 0.1u
DVCC1
GND

+VCC_MSP430
L4 600Ohm
C9 2.2u
C12 0.1u
DVCC3
GND

HF and LF Oscillators

C13 12p
R4 22
HFXOUT
X1
GND
GND
GND
FY0800018
8MHz
C14 12p
HFXIN

C16 12p
LFXOUT
X2
C18 12p
ABS07−32.768KHZ−9−T
32.768kHz
LFXIN
GND

US−subsystem oscillators

GND
C15 12p
R5 22
USSXTOUT
Y1
AWSCR−8.00CV−T
8MHz
C17 12p
USSXTIN

Debug LED

LED_MSP430
D1
2.65V
Green
R6 22
GND

Sheet: /MSP/
File: msp.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 9.0.5 | Id: 2/8

Pre-Amplifier

+V_PREAMP
C23 2.2u
C24 0.1u
GND
C25 10nF
R7 2k
GND
C26 1.5p
R8 2k
R10: to limit current when C charges up
PREAMP_SHDN_N
+V_PREAMP
U3 OPA836IDBVT
C28 10nF 100V
R10 100
RX_IN
DC_bias lies here
R12 47k
VGA_REF
C63 2.2u
GND
VGA_IN
to keep pramp save from direct bias, first enable preamp, then LT3463
PREAMP_SHDN_N
R14 10k
PREAMP_PWR_END
GND
TP5 preamp_pwr_en
U6 TPS22916CYFPR
ON VIN VOUT GND B2 B1 A2 A1
C32 2.2u
V_RX 3,7
+V_PREAMP

VGA

+V_VGA
C19 2.2u
C20 0.1u
GND
U2 AD8338ACPZ-R7
VBAT VREF VAGC INMD INMR INPD INPR MODE OFSN OUTM OUTP DETO FBKM FBKP GAIN EP COMM
VGA_REF
C27 1n
C30 1n
VGA_REF
vga_vref TP2
GND
VGA_OUT
R11 400
VGA_GAIN
GND
TP4 vga_pwr_en
U5 TPS22916CYFPR
VGA_PWR_END
C31 2.2u
V_RX 3,7
+V_VGA
GND

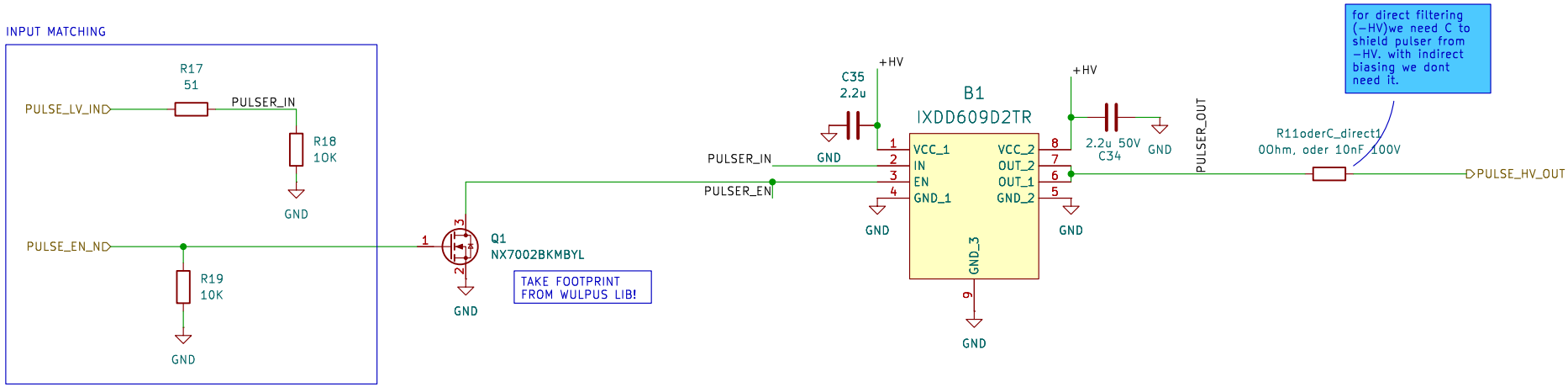
Output Buffer + LPF @4MHz

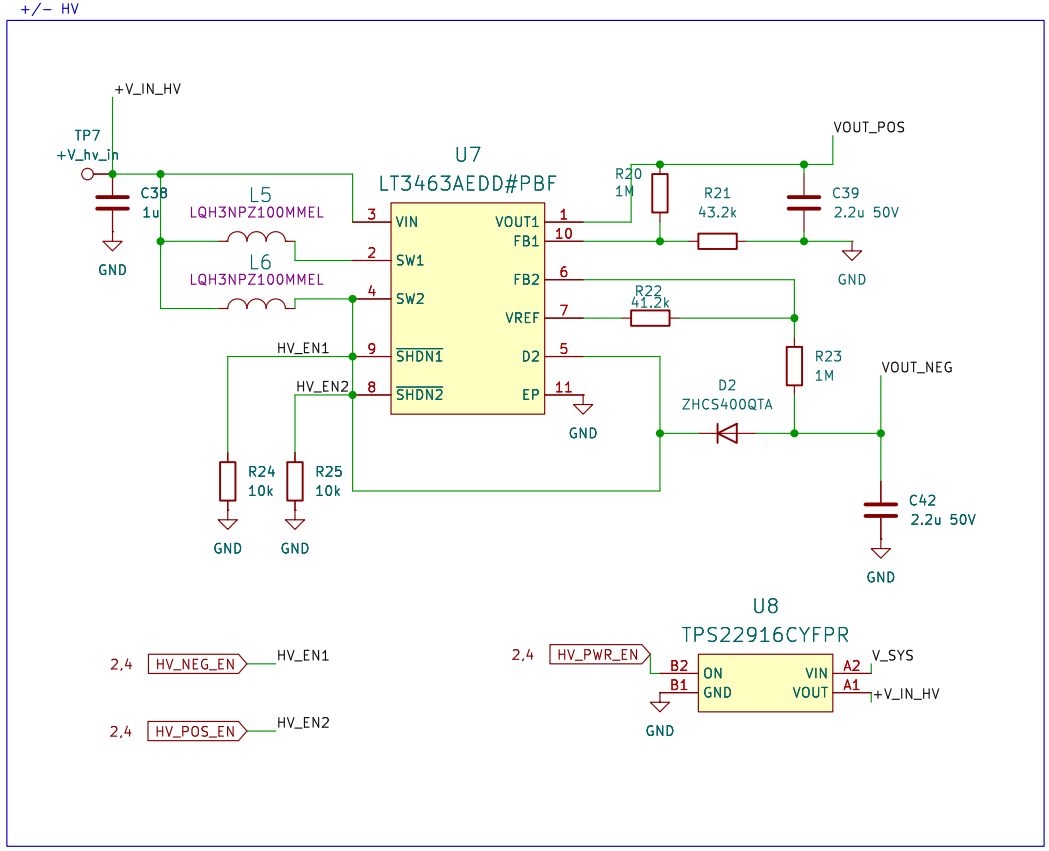
V_RX 3,7
C21 2.2u
C22 0.1u
GND
Spannungsfolger trennt Signal vom RX-path
FILTER_SHDN_N
U4 OPA836IDBVT
BUF_OUT
R9 400
SIG_FILT_IN
C29 100p
GND
FILTER_SHDN_N
R13 10k
LP_FILT_END
GND
filt_shdn_en TP3

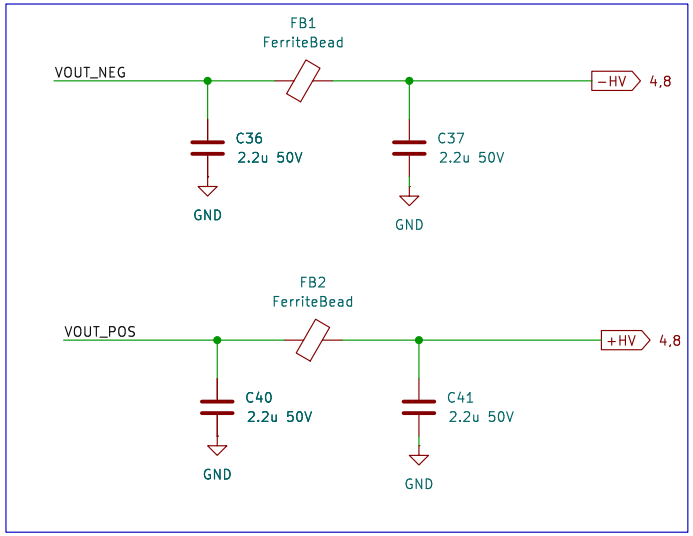
VGA_CONTROL

TP6 TestPoint
R15 1.3k
VGA_GAIN_RC_SINKD
VGA_GAIN
C33 1n
GND
IC1 TPL0501-100RSER
R16 2.7k
VGA_GAIN_RC_END
VGA_GAIN_CSD
H DIN CS VDD W SCLK GND
V_RX 3,7
VGA_SPI.MOSI
RX_PATH_SPI.CLK
VGA_SPI{CLK MOSI MISO} 2,3
GND

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Size: A3	Date:
KiCad E.D.A. 9.0.5	Rev:
	Id: 3/8







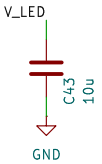
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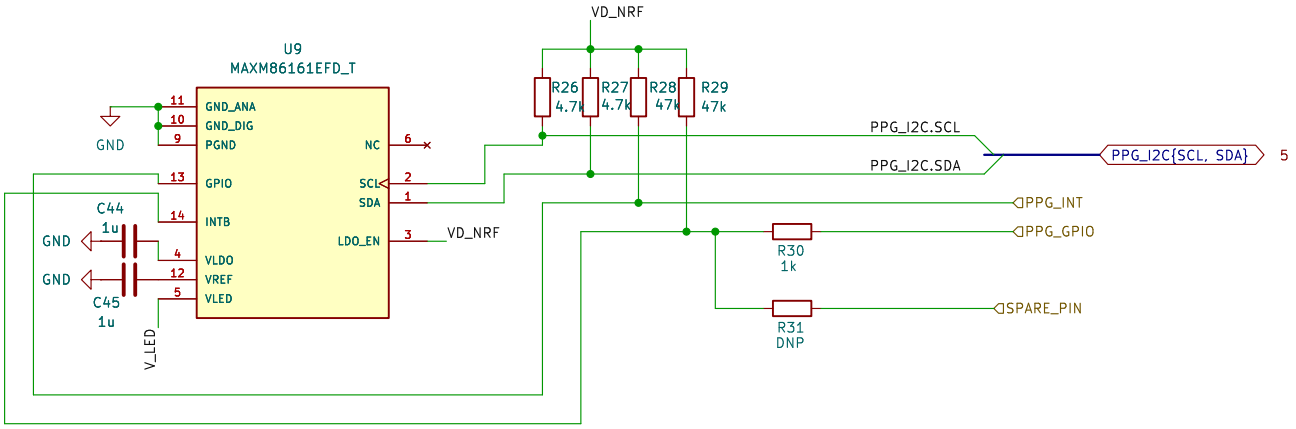
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.5		Id: 4/8

layout: PPG direkt hinter CMUT.
Alternativen mitdenken

VLED based on system application and LED configs used.
V_LED=4.4V

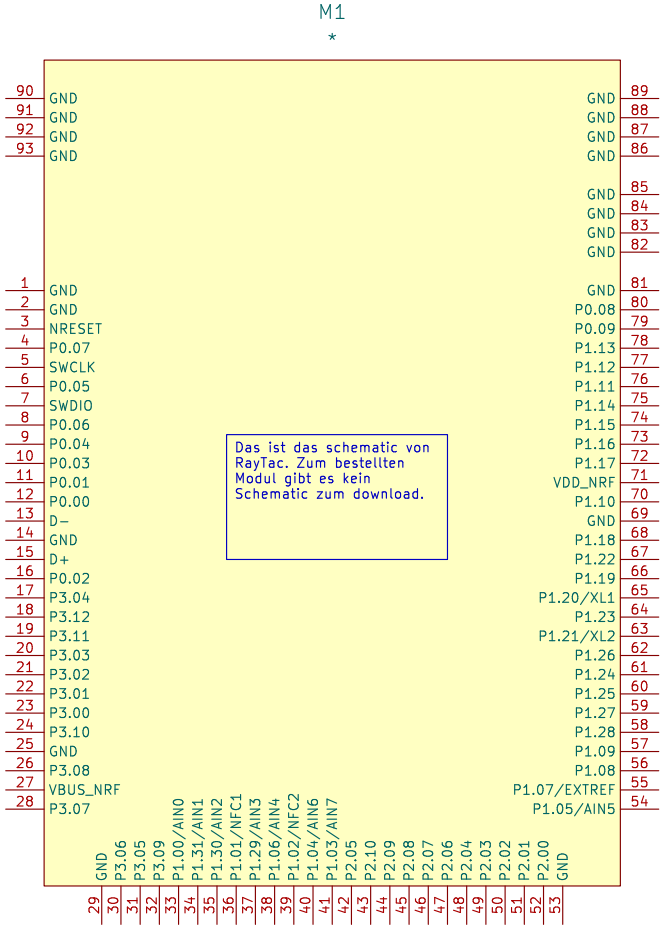




nRF as central clock. it triggers the MSP and the MAXM at the same time, the timedelay until it measures is constant and can then be calibrated.

Alternatively connect it to MSP spare pin and trigger MAXM when pulse is triggered. time delay will be within microseconds

Sheet: /PPG/		
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.5		Id: 5/8

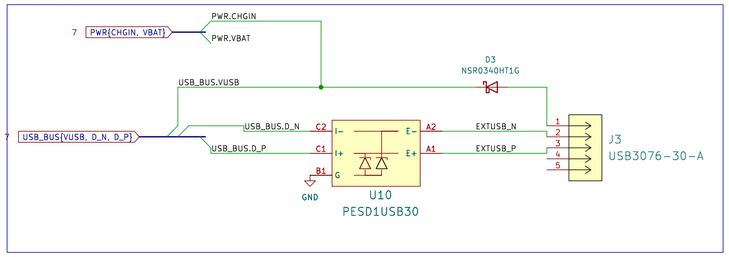


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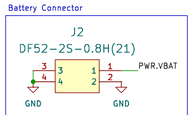
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.5		Id: 6/8

MICRO USB CONNECTOR



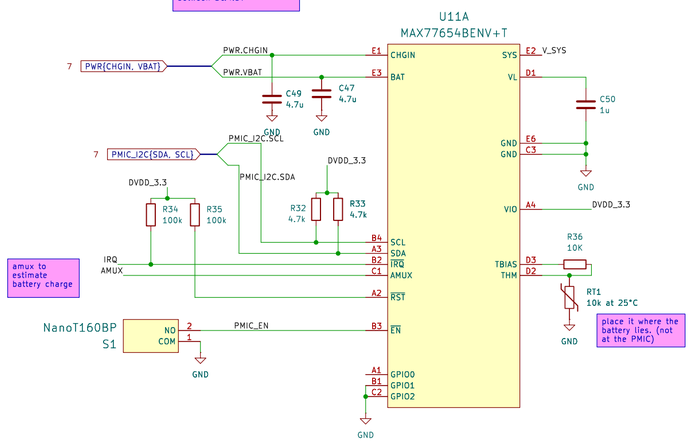
Battery Connector

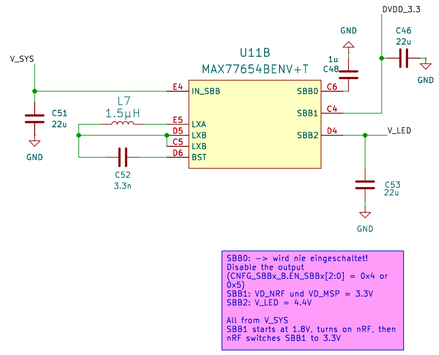


HV POWER IS ON TX PAGE

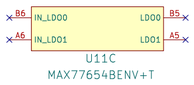
PMIC

V_SYS is dep. on Battery or charging, somewhere between 3&4.5V

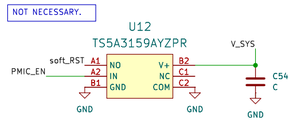




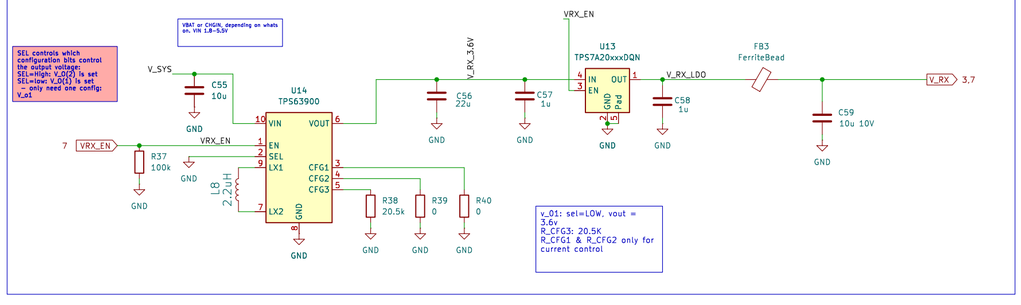
LDO wird nicht benötigt, eighter leave all pins unconnected or connected to ground

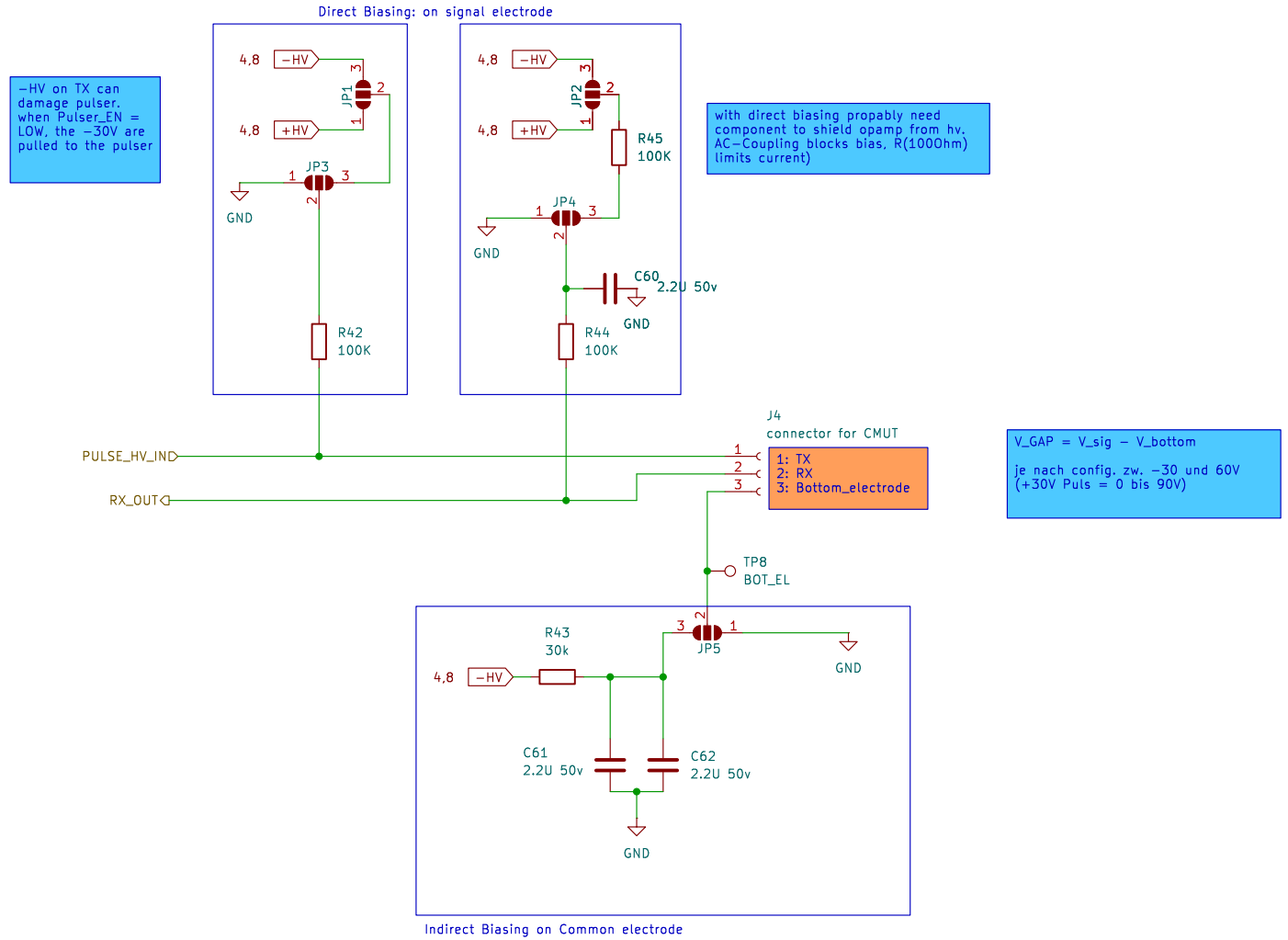


NOT NECESSARY.



to generate V_RX rail: TPS63900 generates 3.6V from V_sys and TPS7A20 LDO cleans it to 3.3V





Sheet: /CMUT/		
File: cmut.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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